

Object-Oriented Programming Assignment

Topics: Classes and Objects, Constructors, Inheritance, Polymorphism, Abstraction, Encapsulation.

1. Classes and Objects:

- Create a class 'Student' with attributes name, rollNo, and marks.
- Create objects and display their details.

2. Constructors:

- Add a constructor to initialize student details.
- Create multiple student objects using different constructors.

3. Inheritance:

- Create a base class 'Person' and derived class 'Teacher'.
- Show inheritance with additional attributes.

4. Polymorphism:

- Create method 'displayInfo()' in both Student and Teacher classes.
- Demonstrate method overriding.

5. Abstraction and Encapsulation:

- Create an abstract class 'Shape' with abstract method area().
- Implement subclasses Circle and Rectangle.
- Use private variables with getter/setter methods.